Propuesta de Software Planning and UML

Grupo Prolecom

January 11, 2026

1 Risk management, product and sprint backlogs and scheduling

1.1 Risk management

In this section, we will identify, quantify, and classify the various risks that may arise during the software development process. Additionally, we will provide a detailed assessment of the likelihood of occurrence, the potential impact of each risk, and the corresponding protocols to be followed in the event they materialize.

Probability and Impact Definitions

Description	**Probability Range**
Not Probable: The event is highly unlikely to occur.	0% - 20%
Low Probability: The event is unlikely but possible.	21% - 40%
Moderate Probability: The event has an even chance of occurring.	41% - 60%
High Probability: The event is likely to occur.	61% - 80%
Very High Probability: The event is almost certain to occur.	81% - 100%

Table 1: Probability of Occurrence

Impact Level	**Description**
Low Impact	Minimal effect on the project. No significant changes required.
Moderate Impact	Some delays or adjustments needed but manageable within the team.
High Impact	Significant disruptions, requiring immediate attention and resource allocation.
Critical Impact	Severe consequences on project delivery, with major delays or failure possible.

Table 2: Impact Levels

The following table outlines the identified risks associated with the project, including their probability of occurrence, potential impact, and the corresponding action protocol.

Id	Name	Probability	Impact	Action Protocol
001	Scope Creep regarding Visual IDE Functionality	Very High Probability	Critical Impact	Strictly adhere to the requirement of a "Visual-Only IDE". Mitigate by rejecting any attempts to implement code execution backends unless the core MVP is 100% complete.
002	Frontend-Backend Integration Issues (CORS/API)	High Probability	High Impact	Establish a clear API contract (JSON structure) between the ReactJS team and the PHP team before implementation begins. Use tools like Postman to validate headers and responses early.
003	Undefined Deployment Environment	Moderate Probability	High Impact	Since the deployment is "to be defined by client", the team will containerize the application using Docker to ensure portability to any server (AWS, Azure, or On-Premise) without last-minute configuration issues.
004	Difficulty simulating the "Programming Challenges" logic	Moderate Probability	High Impact	Reduce probability by researching existing open-source challenge UI libraries (like CodeMirror) early in the Sprint to ensure the visual aspect mimics HackerRank effectively without needing the heavy backend logic.
005	Database Schema Volatility due to Role Hierarchy	Low Probability	Moderate Impact	Accept the risk but mitigate it by finalizing the Entity-Relationship diagram for the 5 distinct roles (Student, Prof, TA, Mod, Admin) in the first Sprint to prevent schema refactoring later.

Id	Name	Probability	Impact	Action Protocol
006	Need for team training in ReactJS/PHP	High Probability	High Impact	Provide immediate internal workshops or share specific documentation/tutorials on the chosen stack (React Hooks, PHP PDO) to minimize delays caused by learning curves during development.
007	Changes in requirements after development completion	High Probability	High Impact	Establish a strict communication protocol to clarify that no new functional requirements will be accepted after the design phase is finalized, unless approved through a change request process.
008	Academic Integrity Vulnerabilities in Forum	Low Probability	Critical Impact	Reduce probability by implementing strict moderation tools for the "Moderator" role and automatic keyword filtering to prevent students from sharing full solution code in the Q&A section.

Table 3: Identified Risks and Action Protocols

1.2 Product Backlog

ID	Priority	Depend.	Item	Est. (h)
PB1	0	None	**Project Setup & Research:** Investigation of the development stack (PHP/React), environment setup, and repository initialization. Includes feasibility evaluation of the IDE integration.	6
PB2	0	None	**Database Schema Definition:** Design the database schema covering users (Students, Professors, TAs, Mods, Admin), courses, challenges, and forum entities. Includes ER diagram and SQL script generation.	8
PB3	1	PB1, PB2	**(US-STU-01)** As a Student, I want to register and log in using a username and password, so that I can access the platform and personalize my learning experience.	6

ID	Priority	Depend.	Item	Est. (h)
PB4	1	PB3	**(US-PROF-02)** As a Professor, I want to be able to create new public courses (defining Title, Description, Language), so that I can share new subjects and expand the community's educational offering.	8
PB5	1	PB3	**(US-SUP-02)** As a Support Staff member, I want to be able to assign and update user roles (Student, Professor, etc.), so that I can manage user permissions and access levels.	4
PB6	1	PB4	**(US-STU-03)** As a Student, I want to be able to enroll in one or more courses, so that I can access their specific content and challenges. Includes confirmation message and dashboard update.	6
PB7	1	PB6	**(US-STU-02)** As a Student, I want the system to display a main page with easy navigation between available courses (cards with descriptions), so that I can quickly find the material I need.	6
PB8	1	PB4	**(US-PROF-03)** As a Professor, I want to create, edit, and delete programming exercises (defining difficulty, problem statement, test cases), so that I can design custom practice challenges for my students.	12
PB9	1	PB8	**(US-STU-09)** As a Student, I want to view and write code for challenges in the visual IDE (supporting Python syntax), so that I can focus on solving the problem logic without needing local installations.	14
PB10	2	PB9	**(US-STU-08)** As a Student, I want to access a bank of programming exercises organized by difficulty (Easy, Medium, Hard), so that I can practice progressively.	6
PB11	2	PB9	**(US-STU-10)** As a Student, I want to view my basic progress and results (attempted vs completed challenges) for the challenges I complete, so that I can monitor my learning curve.	5
PB12	2	PB8	**(US-PROF-06 / US-TA-05)** As a Professor or TA, I want to review and provide feedback on the programming challenges submitted by students, so that I can ensure the quality and instructional clarity of the exercises.	10

ID	Priority	Depend.	Item	Est. (h)
PB13	2	PB4	**(US-PROF-01 / US-TA-03)** As a Professor or TA, I want to upload, edit, and organize learning materials (PDFs, video links) by topic, so that students have structured and up-to-date resources.	8
PB14	2	PB13	**(US-STU-07)** As a Student, I want to be able to review learning material uploaded by professors (view in browser and download PDFs), so that I can study and reinforce my concepts.	6
PB15	2	PB6	**(US-STU-04 / US-STU-05)** As a Student, I want to post questions in the course forum, categorize them, and reply to others, so that I can collaborate and help my peers.	10
PB16	3	PB15	**(US-PROF-04 / US-TA-01)** As a Professor or TA, I want to be able to answer student questions in the course forum with a visual highlight (tag), so that I can provide official academic guidance.	4
PB17	3	PB15	**(US-STU-06)** As a Student, I want to be able to report posts that violate platform rules, so that I help maintain academic honesty and a respectful environment.	4
PB18	3	PB17	**(US-MOD-01 / US-MOD-02)** As a Moderator, I want to supervise flagged posts via a dashboard and hide/ban content that violates rules, so that I can maintain a safe environment.	8
PB19	3	PB6	**(US-STU-11 / US-STU-12)** As a Student, I want to access quizzes created by the professor and optionally convert questions into flashcards, so that I can test my knowledge.	10
PB20	3	PB4	**(US-PROF-07)** As a Professor, I want to be able to add Teaching Assistants to my course by searching for existing users, so that I can delegate mentorship tasks.	5
PB21	2	PB1	**(US-ADM-01)** As an Administrator, I want the system to display a log of relevant platform activities on a dashboard, so that I can monitor the overall health and growth of the platform.	6
PB22	3	PB3	**(US-SUP-01)** As a Support Staff member, I want to create, modify, and delete user accounts (with confirmation), so that I can perform maintenance and resolve login issues.	6
PB23	4	All	**Testing and Deployment:** Functional testing, Integration testing (especially IDE and Forum), Performance testing (load times), and Deployment to production.	24

ID	Priority	Depend.	Item	Est. (h)

Table 4: Product Backlog of the Project

1.3 Sprint Backlog

PB Item	User Story	Tasks	Assigned To
Sprint 1: Architecture, Authentication & Course Foundations (Weeks 1–3)			
PB1	Research & Setup	- Research development stack (PHP/React). - Initialize GitHub repo & CI/CD pipeline. - Setup Docker environment.	Eimmy Ochoa José Toapanta
PB2	Database Definition	- Design ER Diagram. - Create SQL Scripts / ORM Models. - Define constraints for Users and Courses.	Sebastian Holgin Leonardo
PB3	(US-STU-01) Register & Login	- Implement JWT Authentication backend. - Design Login/Register UI. - Integrate frontend with Auth API.	Issac Maza Eimmy Ochoa
PB4	(US-PROF-02) Create Public Courses	- Create CRUD endpoints for Courses. - Design "Create Course" form. - Implement course visibility logic.	José Toapanta Leonardo
PB5	(US-SUP-02) Manage Roles	- Implement Role-Based Access Control (RBAC). - Create admin interface for role assignment.	Sebastian Holgin
Sprint 2: Enrollment, Content Management & Challenge Backend (Weeks 4–6)			
PB6	(US-STU-03) Enrollment	- Create Student–Course association table. - Implement enrollment validation logic. - Update dashboard UI.	Issac Maza Eimmy Ochoa
PB7	(US-STU-02) Main Page Navigation	- Design course card components. - Implement search and filter logic.	José Toapanta
PB13	(US-PROF-01) Upload Materials	- Configure file storage. - Create UI for PDF/link upload.	Sebastian Holgin Leonardo
PB14	(US-STU-07) Review Materials	- Implement PDF viewer. - Secure download endpoints.	Eimmy Ochoa

PB Item	User Story	Tasks	Assigned To
PB8	(US-PROF-03) Create Exercises	- Design DB schema for challenges and test cases. - Create exercise definition UI. - Validate hidden test cases.	Leonardo Issac Maza
Sprint 3: Visual IDE, Forum & Feedback Loop (Weeks 7–9)			
PB9	(US-STU-09) Visual IDE	- Integrate Monaco Editor. - Implement sandboxed code execution. - Return execution results to UI.	Eimmy Ochoa José Toapanta Sebastian Holgin
PB10	(US-STU-08) Challenge Bank	- Create challenge list view. - Implement difficulty filters.	Issac Maza
PB12	(US-PROF-06) Review & Feedback	- Interface for reviewing student code. - Commenting system on submissions.	Leonardo Eimmy Ochoa
PB15	(US-STU-04) Forum System	- Database design for threads and replies. - UI for posting and categorizing questions.	José Toapanta
PB16	(US-PROF-04) Official Answers	- Highlight professor/TA answers. - Visual distinction for official replies.	Sebastian Holgin
Sprint 4: Assessment, Moderation, Gamification & Final Launch (Weeks 10–12)			
PB19	(US-STU-11) Quizzes & Flashcards	- Implement quiz logic. - Convert questions to flashcards.	Leonardo Issac Maza
PB11	(US-STU-10) Progress Tracking	- Calculate completion percentages. - Visualize progress bars.	Eimmy Ochoa
PB17	(US-STU-06) Reporting System	- Report button on posts. - Backend queue for reports.	José Toapanta
PB18	(US-MOD-01) Moderator Dashboard	- Dashboard for flagged content. - Hide/ban functionality.	Sebastian Holgin
PB23	Testing & Deployment	- Unit & integration testing. - Load testing for IDE. - Production deployment.	All Team Members

Table 5: Sprint Backlog del Proyecto

1.4 Scheduling

ID	Description	Dep	Product Backlog Items	Hours	EST	LFT	Float
A	System Foundation	None	PB1, PB2	14	0	14	0
B	User & Course Management	A	PB3, PB4, PB5, PB6, PB7, PB20, PB22	41	14	55	0
C	Learning Materials Module	B	PB13, PB14	14	55	112	43
D	Challenge & IDE Engine	B	PB8, PB9, PB10, PB12	42	55	97	0
E	Forum & Community	B	PB15, PB16, PB17	18	55	98	25
F	Gamification & Assessment	D	PB11, PB19	15	97	112	0
G	Moderation & Admin Tools	E	PB18, PB21	14	73	112	25
H	Testing & Deployment	C, F, G	PB23	24	112	136	0

Table 6: Activity Arrow Table for Macro-activities

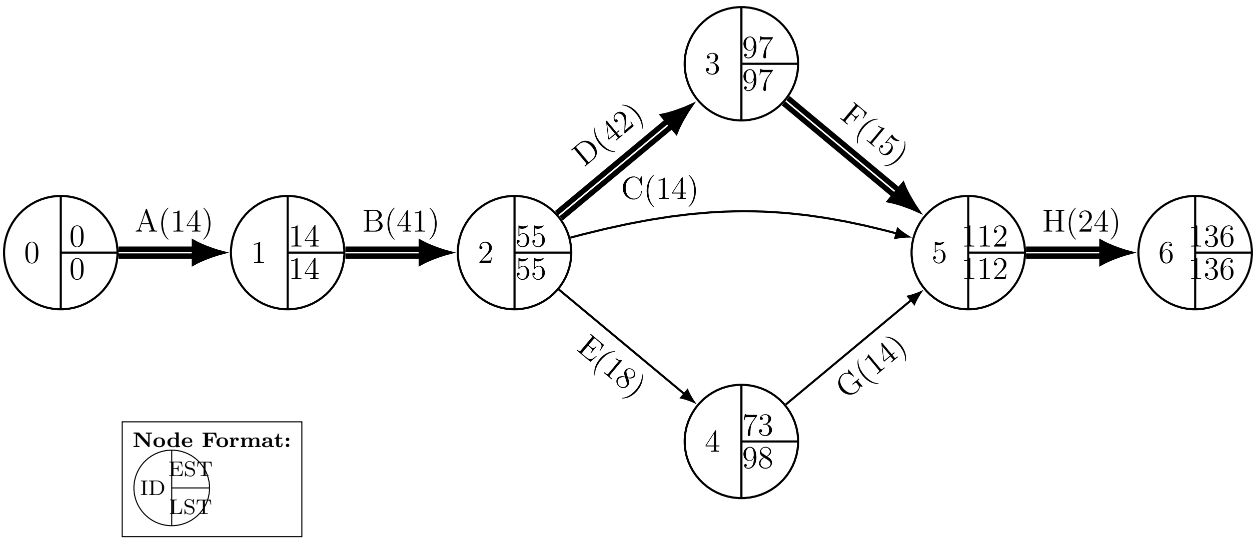

Figure 1: Activity Arrow Diagram of the Project

2 Static UML

2.1 Use Cases - Web Module

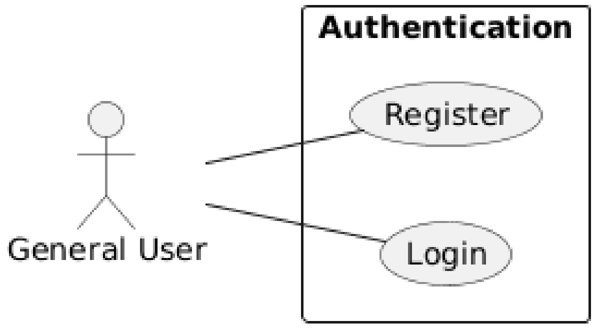

Figure 2: General User Use Cases

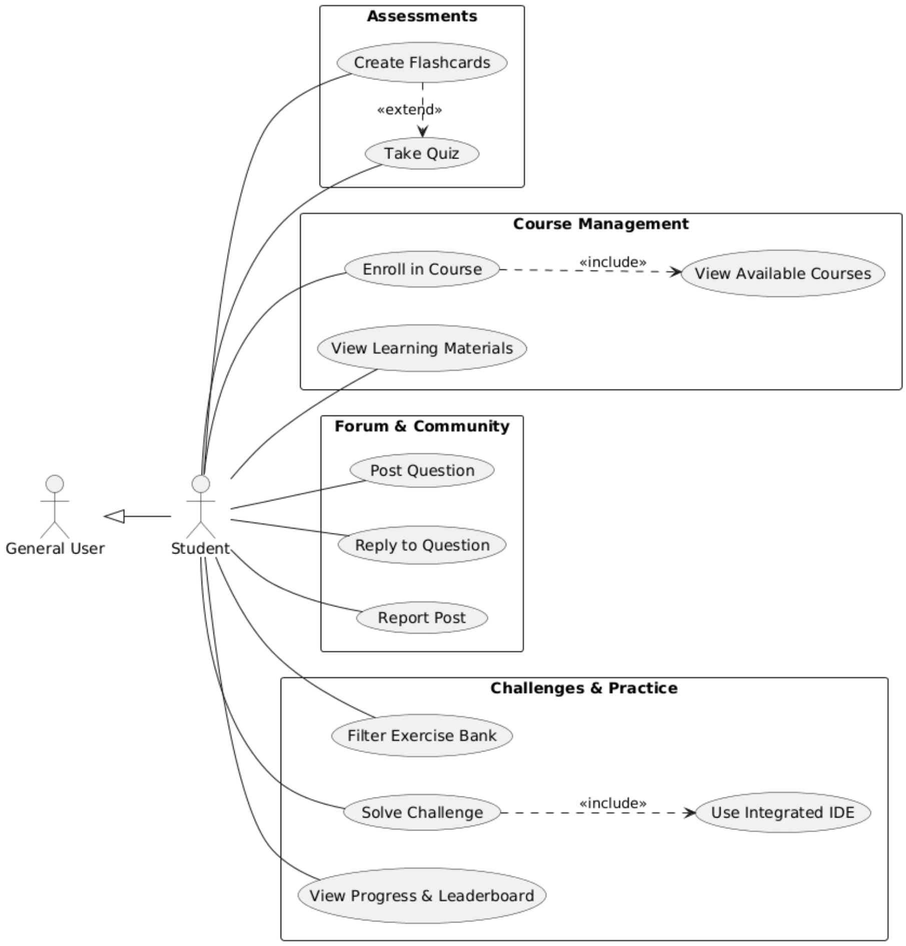

Figure 3: Student Use Cases

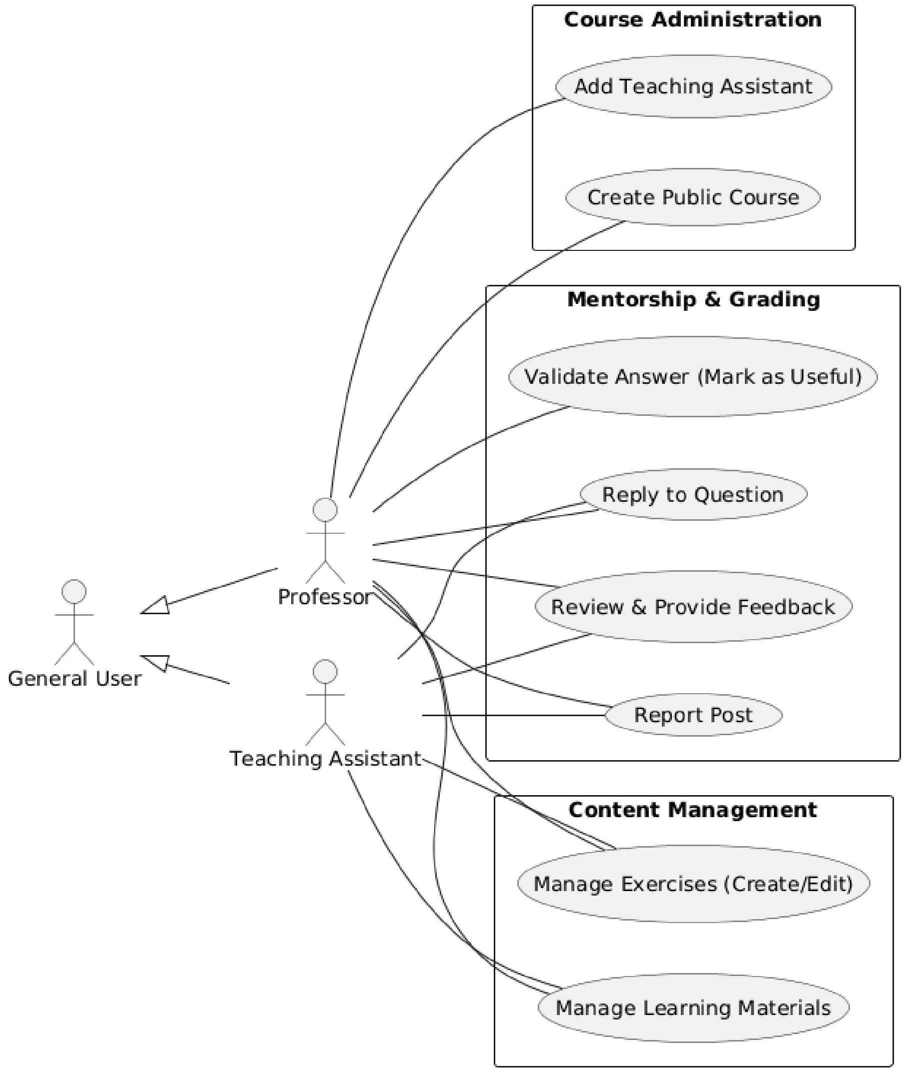

Figure 4: Professor and Teaching Assistant Use Cases

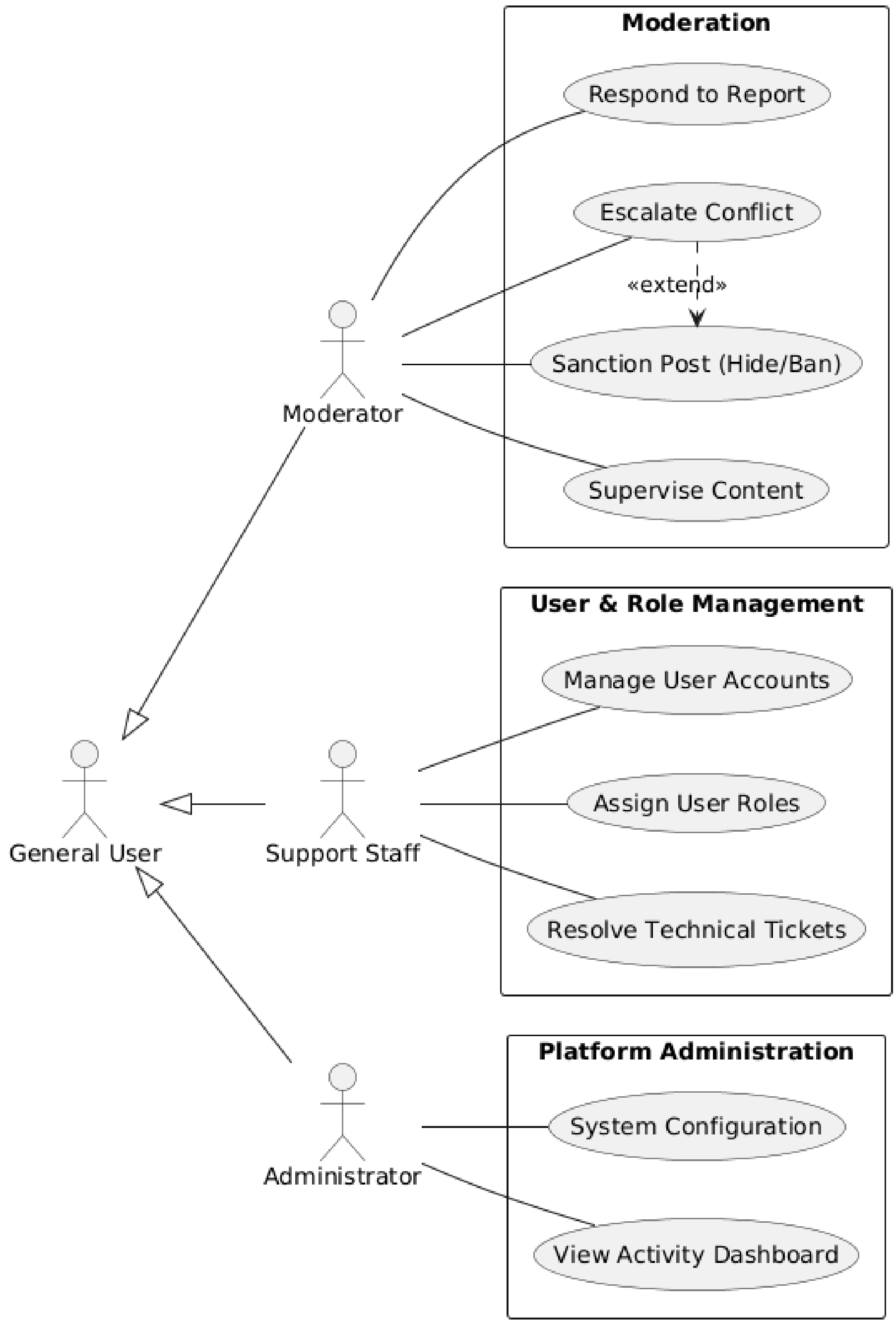

Figure 5: Moderator, Support and Administrator Use Cases

2.2 Use Cases - Mobile Module

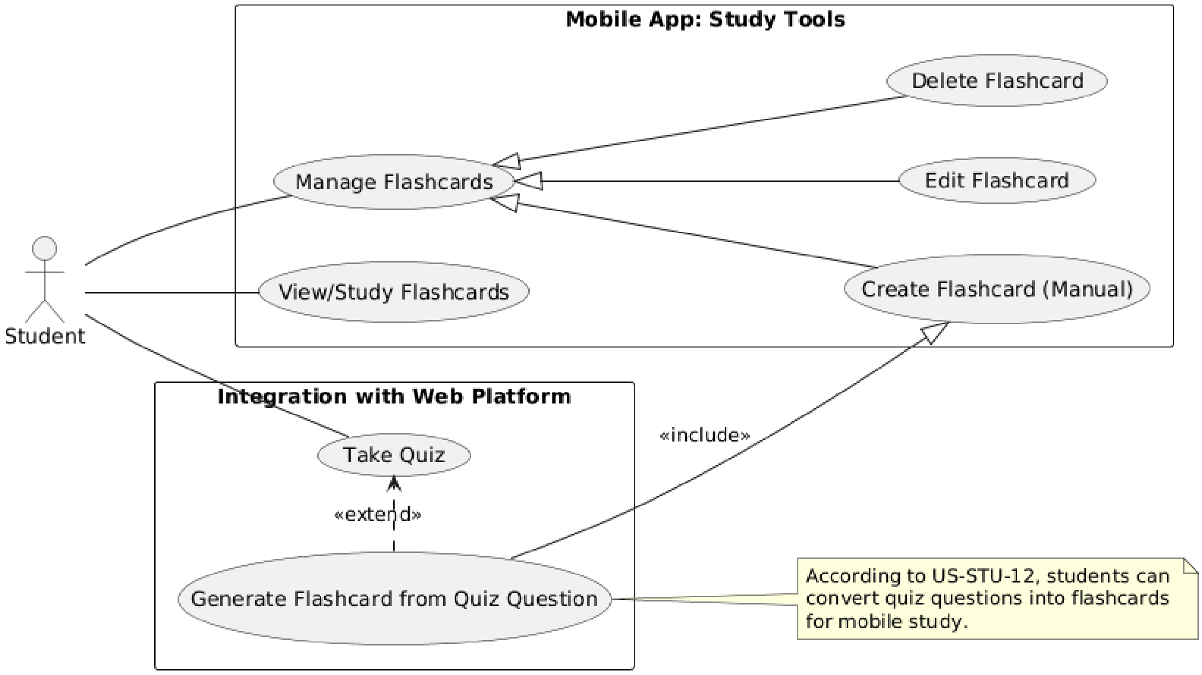

Figure 6: Moderator, Support and Administrator Use Cases

2.3 Class Diagrams - Web Module

2.3.1 Package UserStaff

Description: With business logic of users and account management for Web Module.

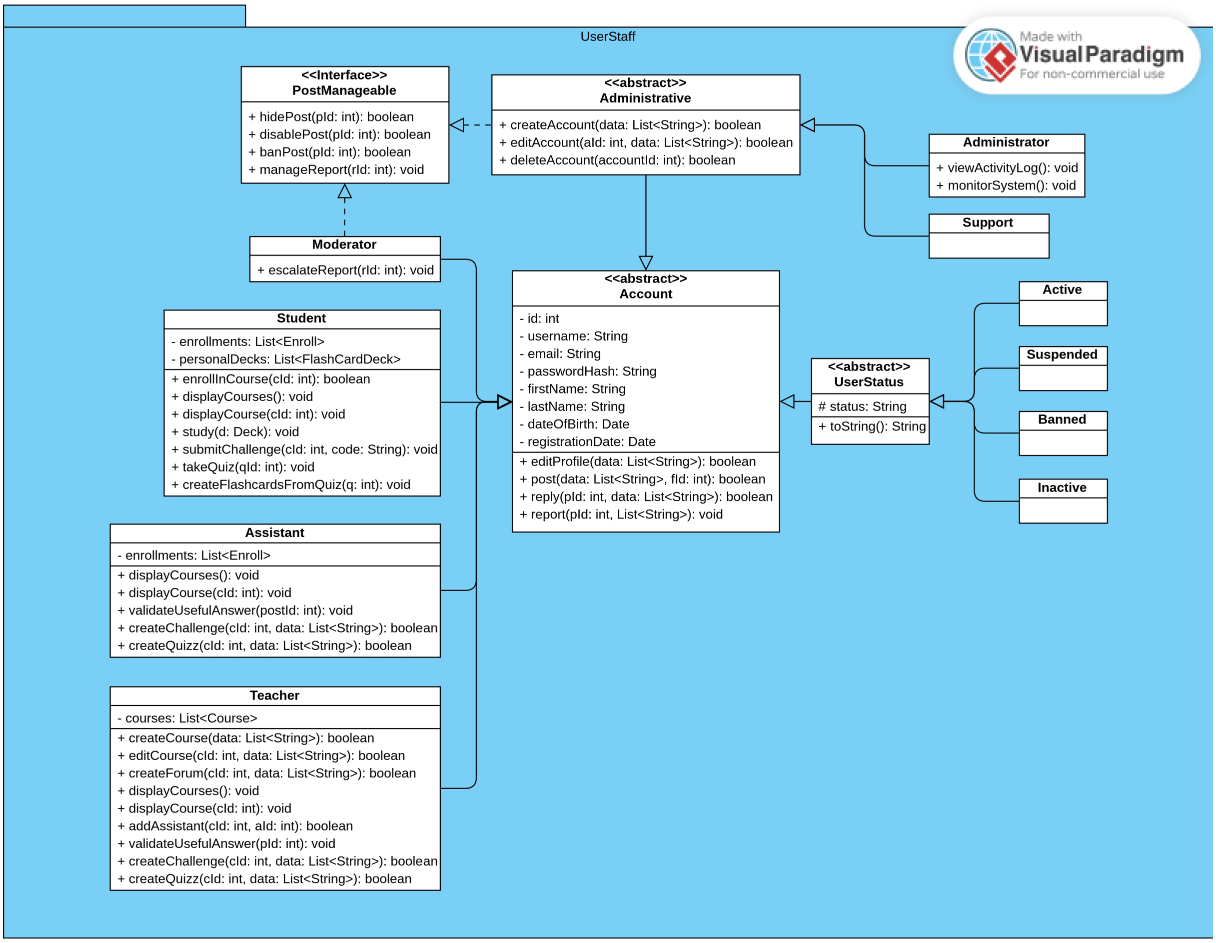

Figure 7: Package UserStaff Class Diagram

2.3.2 Package EnrollmentManager

Description: With business logic of course creation and enrollment processes for Web Module.

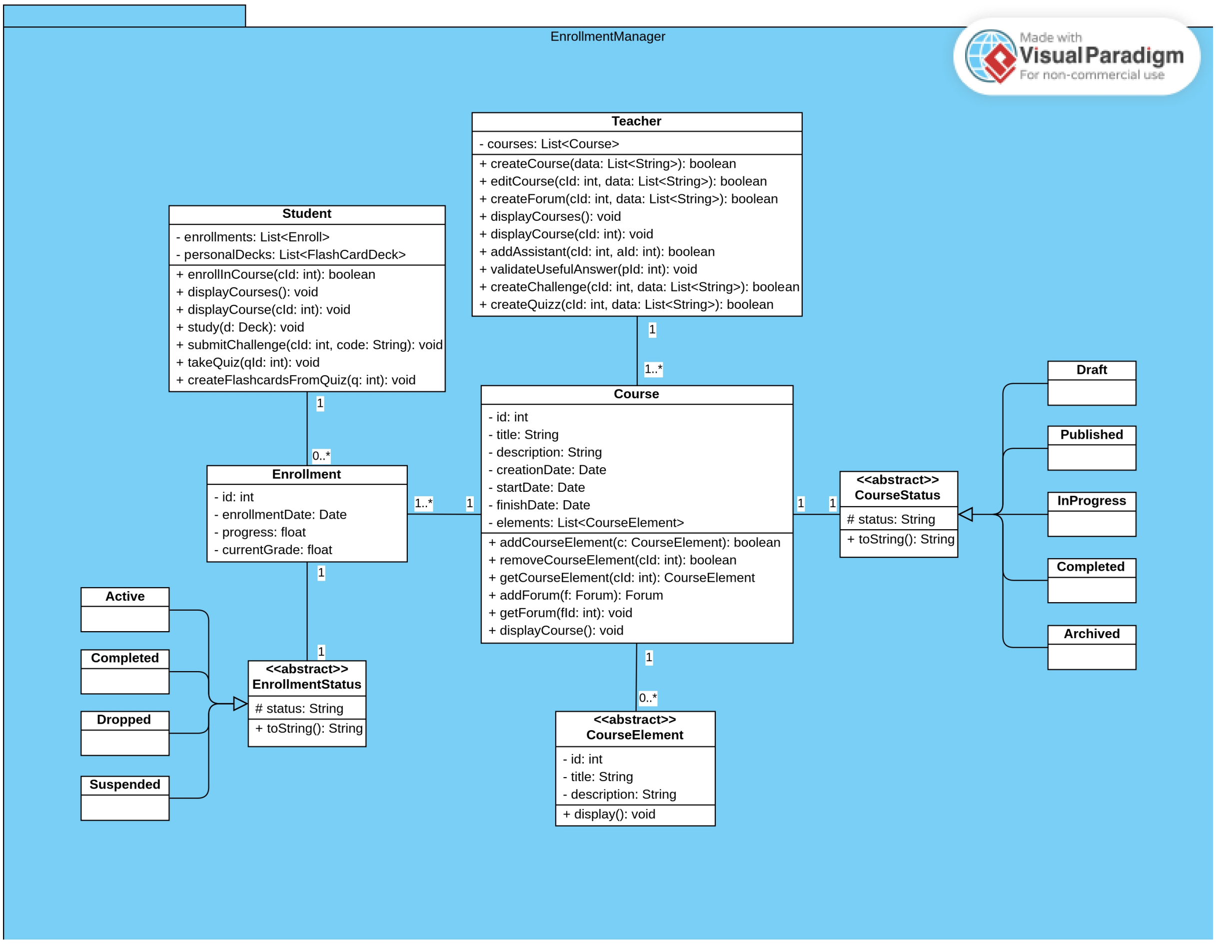

Figure 8: Package EnrollmentManager Class Diagram

2.3.3 Package AcademicContent

Description: With Composite pattern for the hierarchical structure of course elements for Web Module.

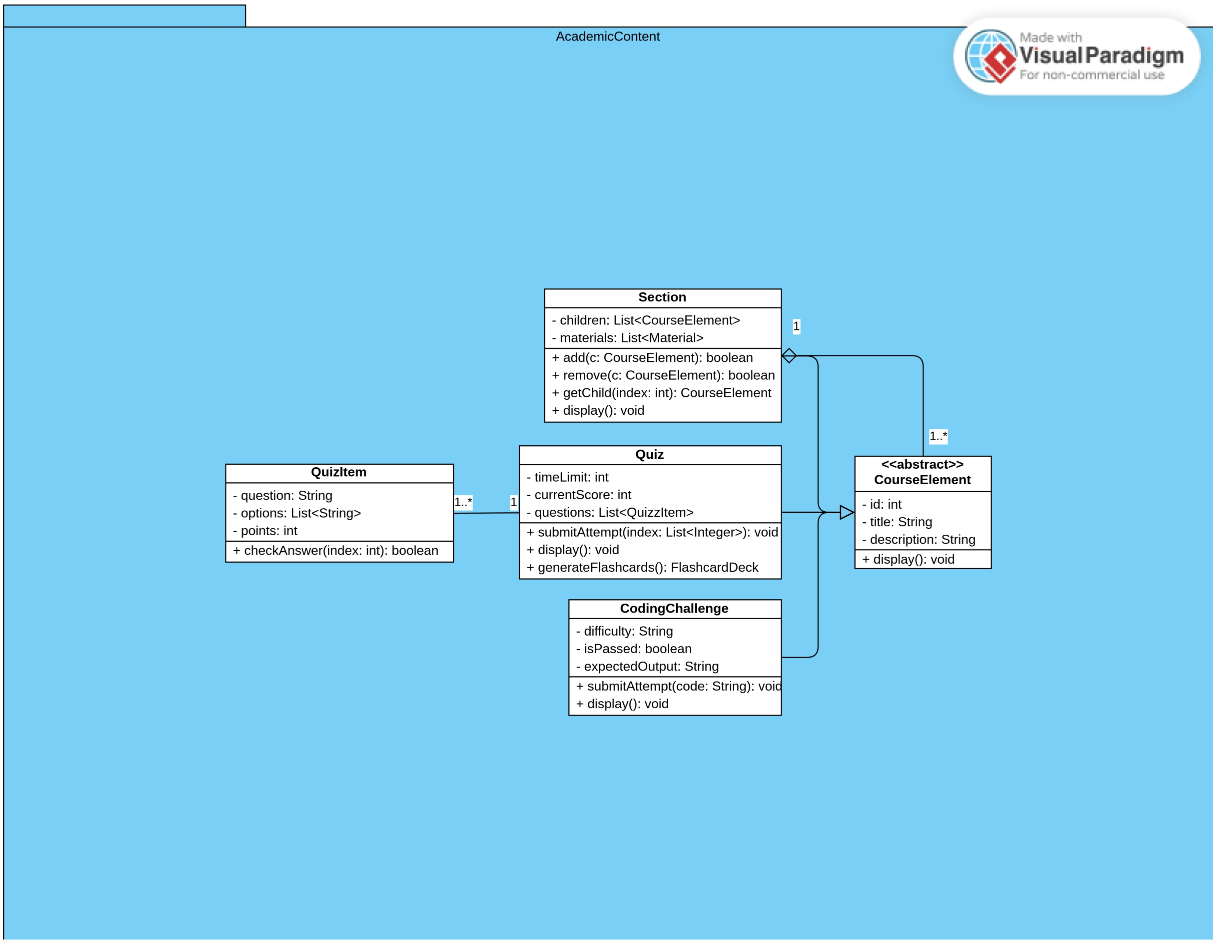

Figure 9: Package AcademicContent Class Diagram

2.3.4 Package MaterialFactory

Description: With Factory Method pattern for dynamic creation of educational resources for Web Module.

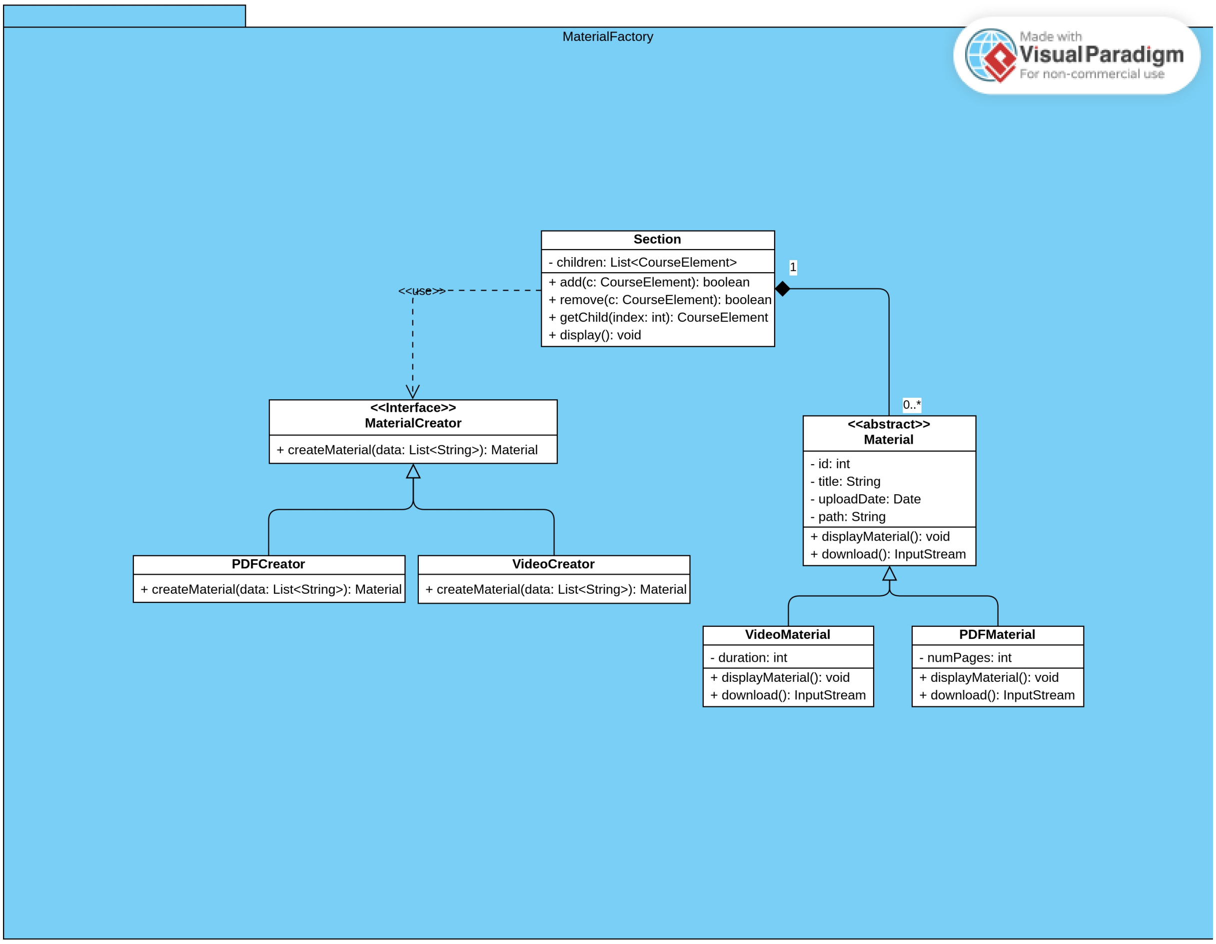

Figure 10: Package MaterialFactory Class Diagram

2.3.5 Package CommunityForum

Description: With Composite pattern for the nested discussion threads for Web Module.

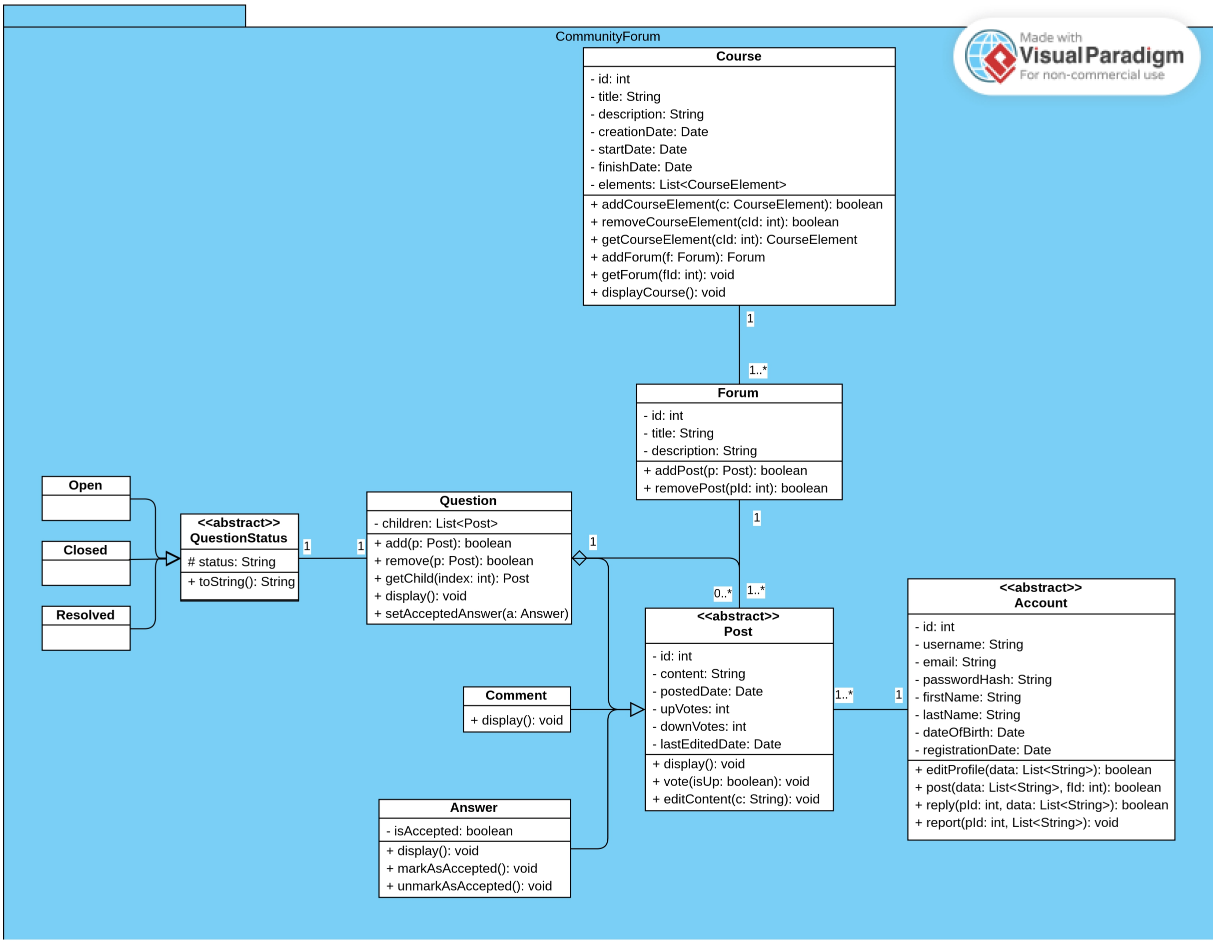

Figure 11: Package CommunityForum Class Diagram

2.3.6 Package ModerationFlow

Description: With Chain of Responsibility pattern for report handling and State pattern for the status for Web Module.

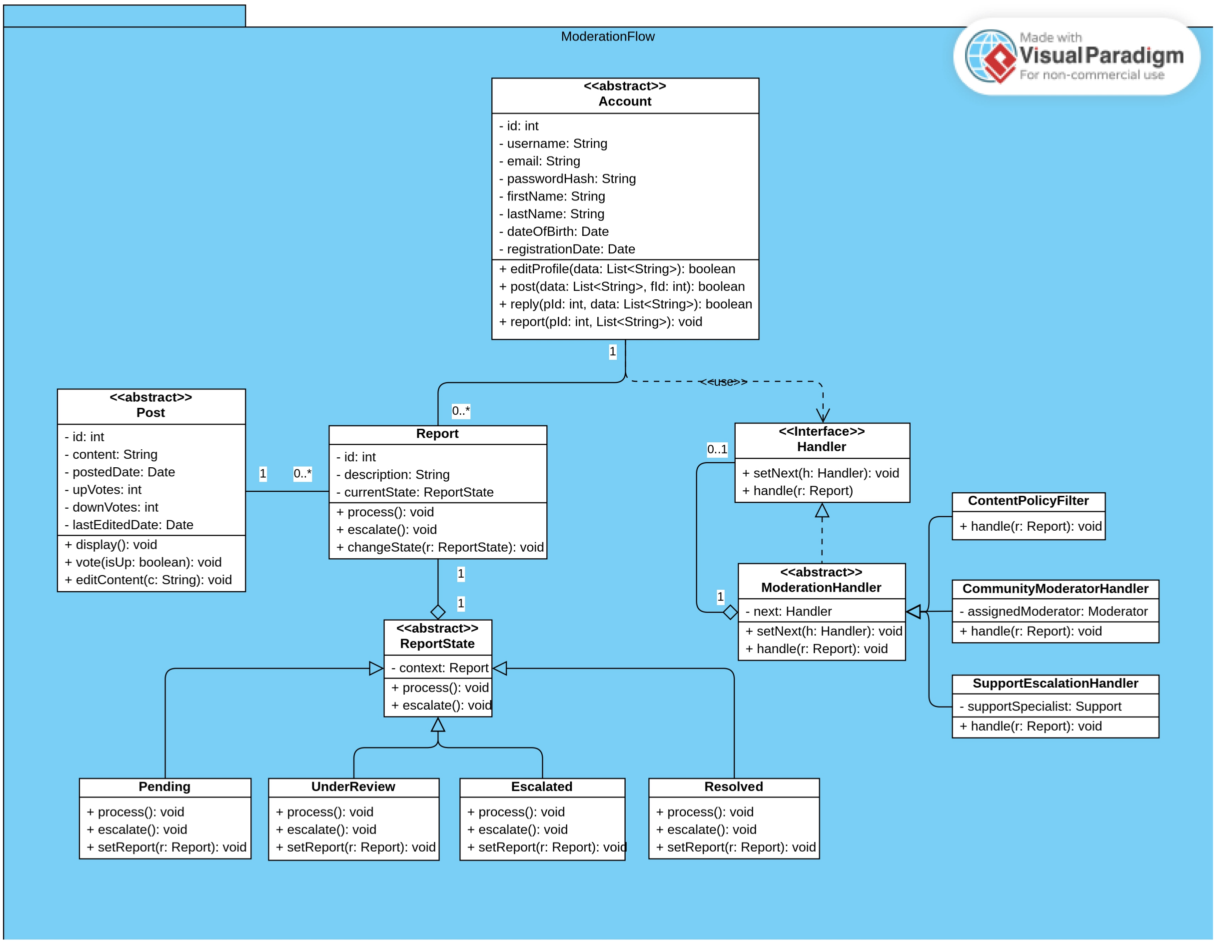

Figure 12: Package ModerationFlow Class Diagram

2.4 Class Diagrams - Mobile Module

2.4.1 Package FlashcardStudy

Description: With Iterator pattern for the navigation of personal study decks for Mobile Module.

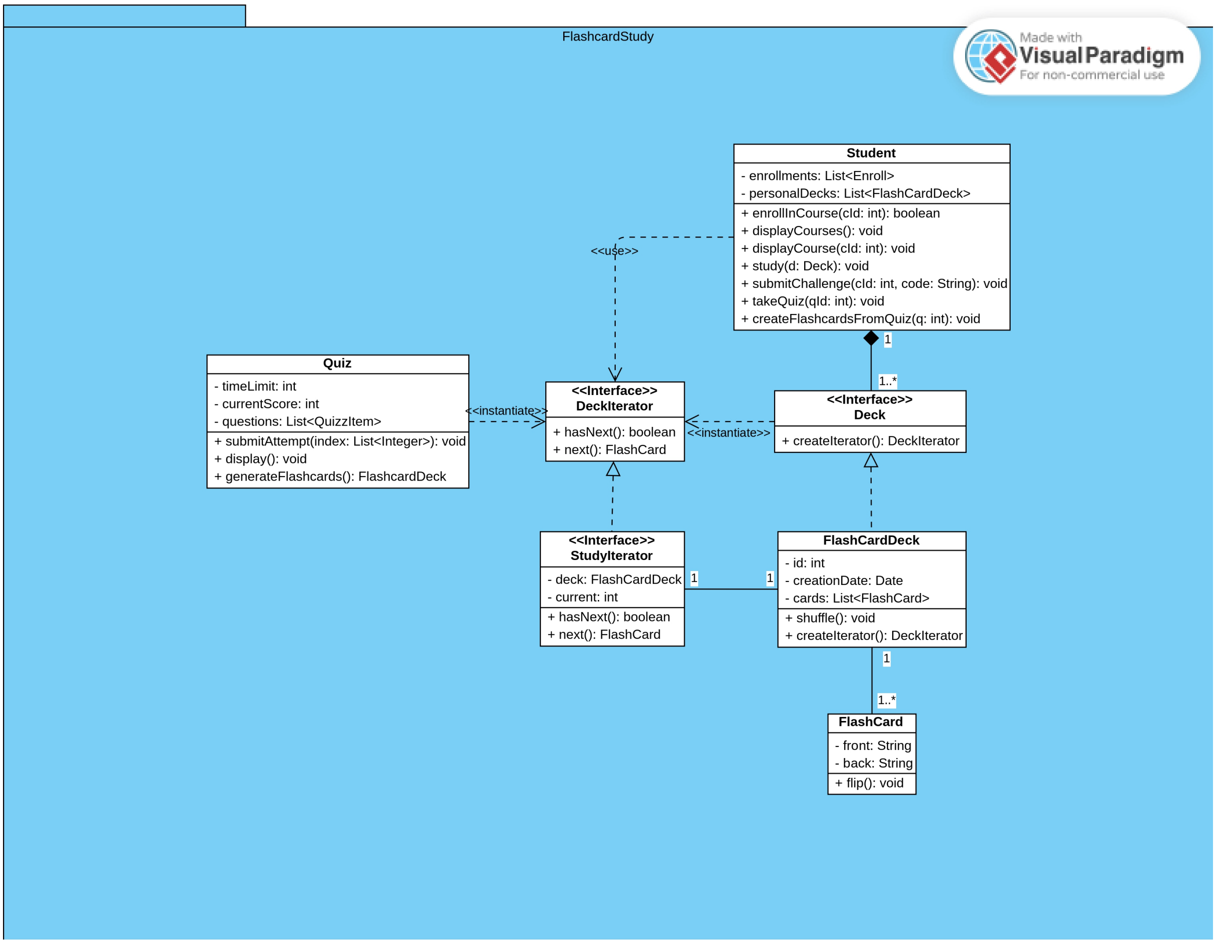

Figure 13: Package FlashcardStudy Class Diagram

2.5 Object Diagrams

2.5.1 Course Actors and Enrollment

Description: Snapshot of Courses, Teachers, Assistants, and Students mediated by Enrollment for Web Module.

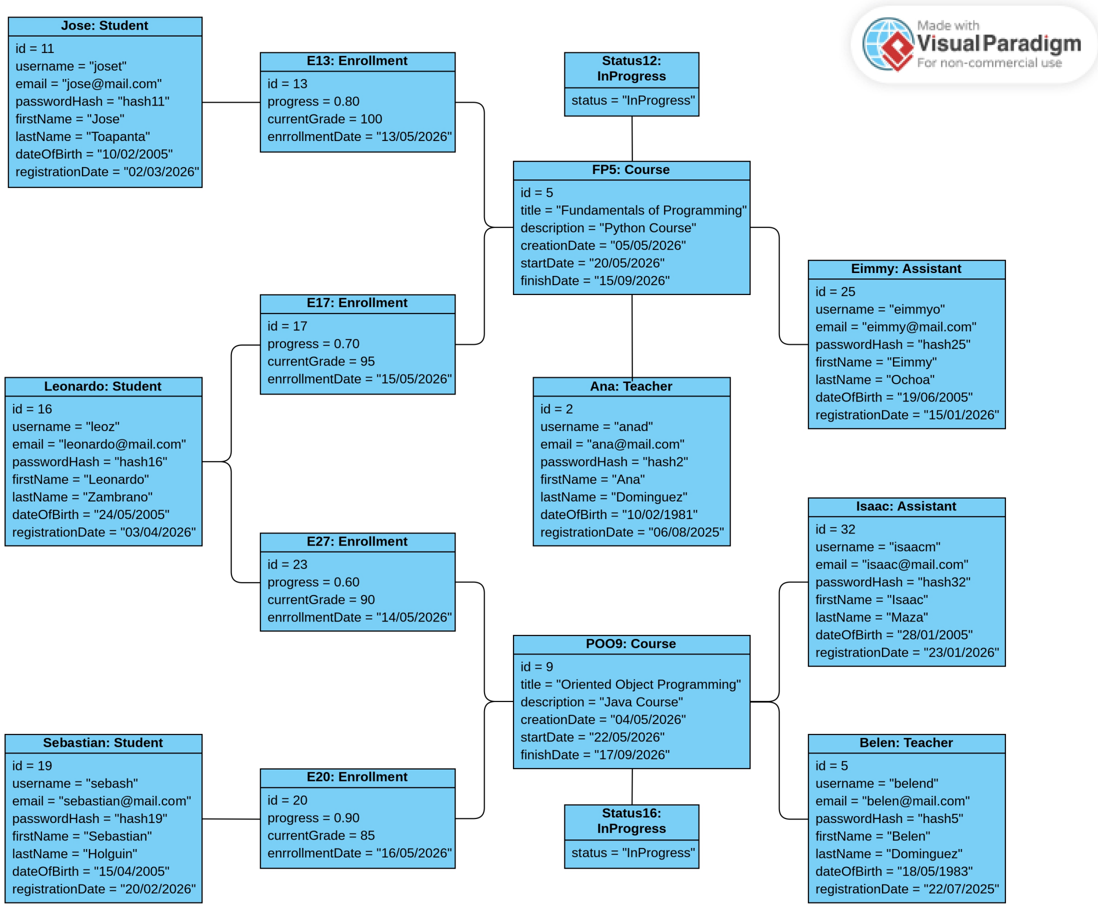

Figure 14: Course Actors and Enrollment Object Diagram

2.5.2 Course Content and Academic Structure

Description: Snapshot of a Course instance with hierarchical Sections and Educational Materials for Web Module.

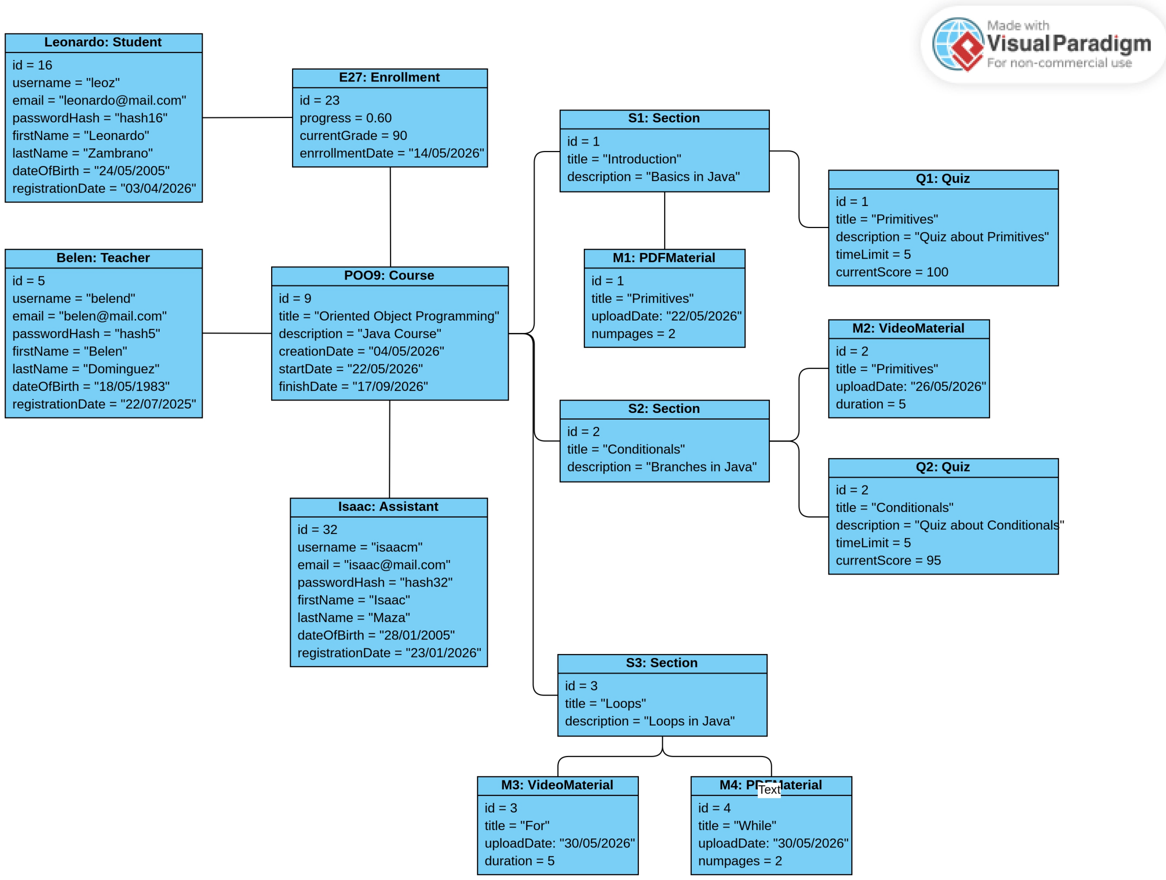

Figure 15: Course Content and Academic Structure Object Diagram

2.5.3 Community Forum and Moderation Flow

Description: Snapshot of Post interactions and the Report escalation process through the moderation chain for Web Module.

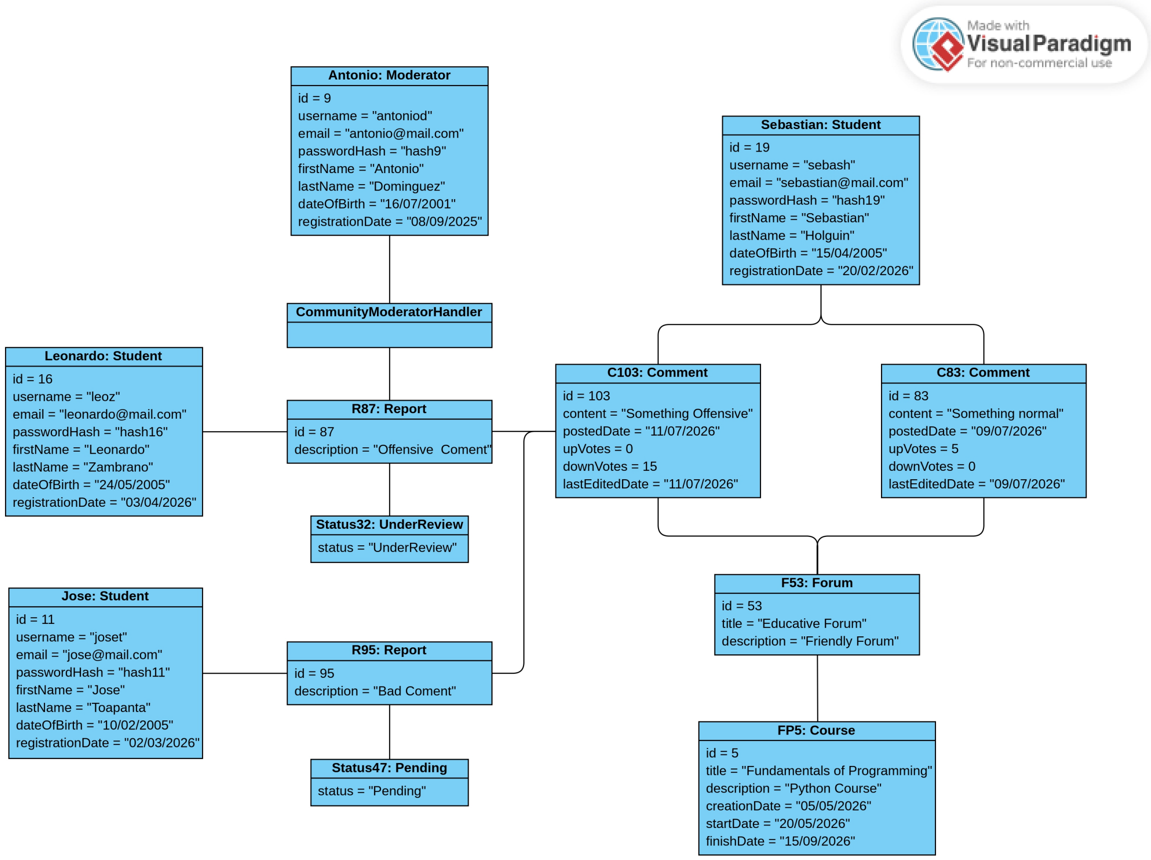

Figure 16: Community Forum and Moderation Flow Object Diagram

2.6 Component Diagrams

2.6.1 BackEnd

Description: High-level component architecture of the Backend System, illustrating the modular separation between authentication, course management, and the challenge engine.

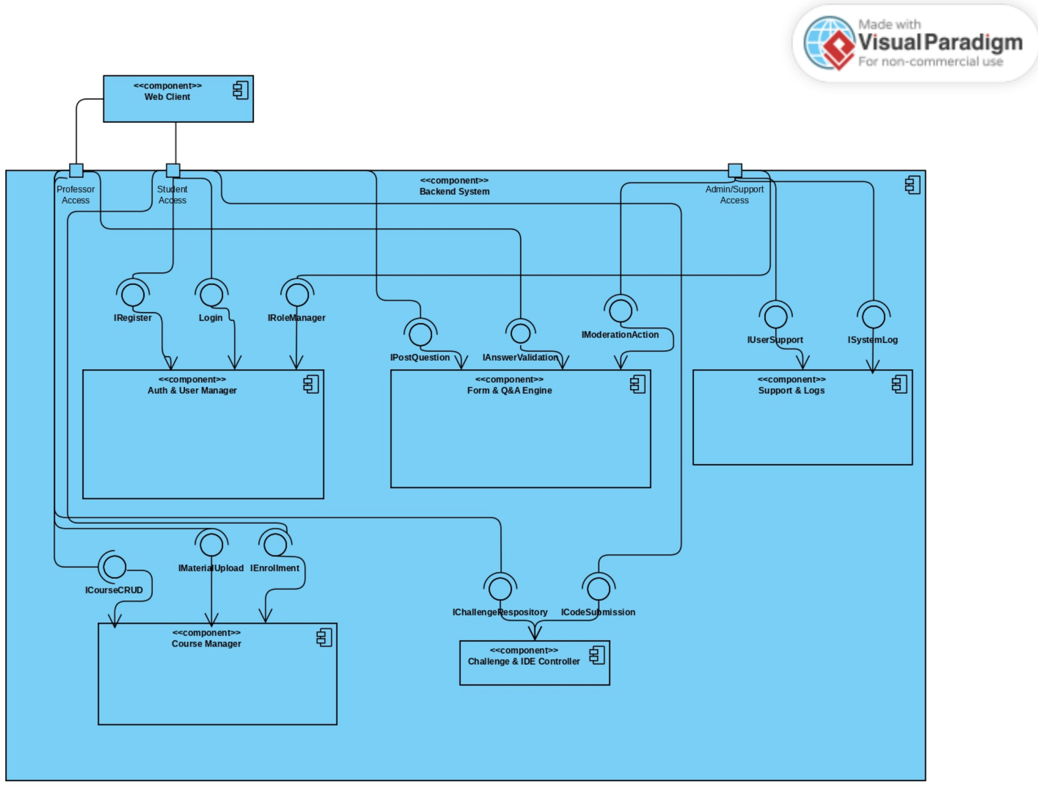

Figure 17: Backend System Component Diagram

2.6.2 FrontEnd

Description: Internal structure of the Web Client (ReactJS), demonstrating the decoupling of UI views from the API Service Layer for external server communication.

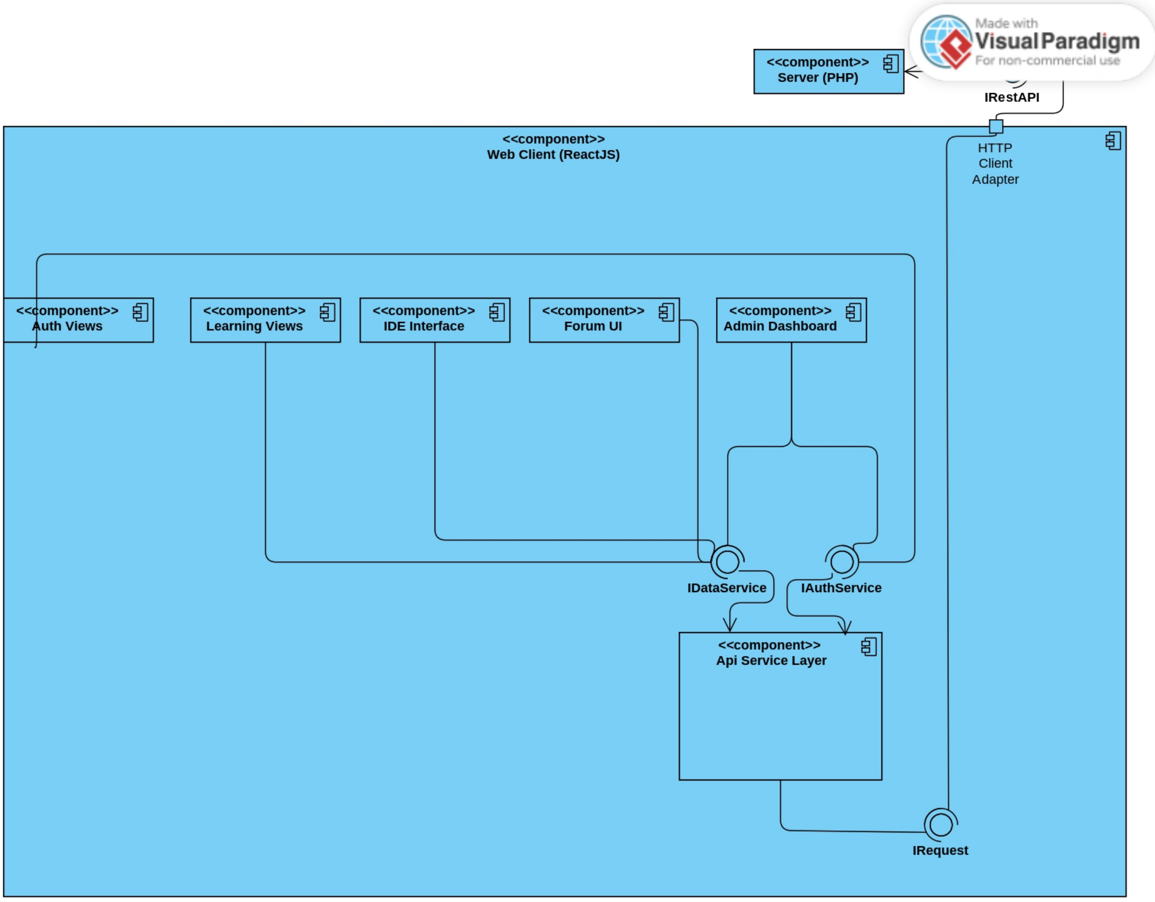

Figure 18: Interface Component Diagram

2.7 Deployment Diagrams

Description: With HTTPS and TCP/IP communication protocols linking the Web Client (ReactJS), PHP Backend Runtime, and MySQL Database Server.

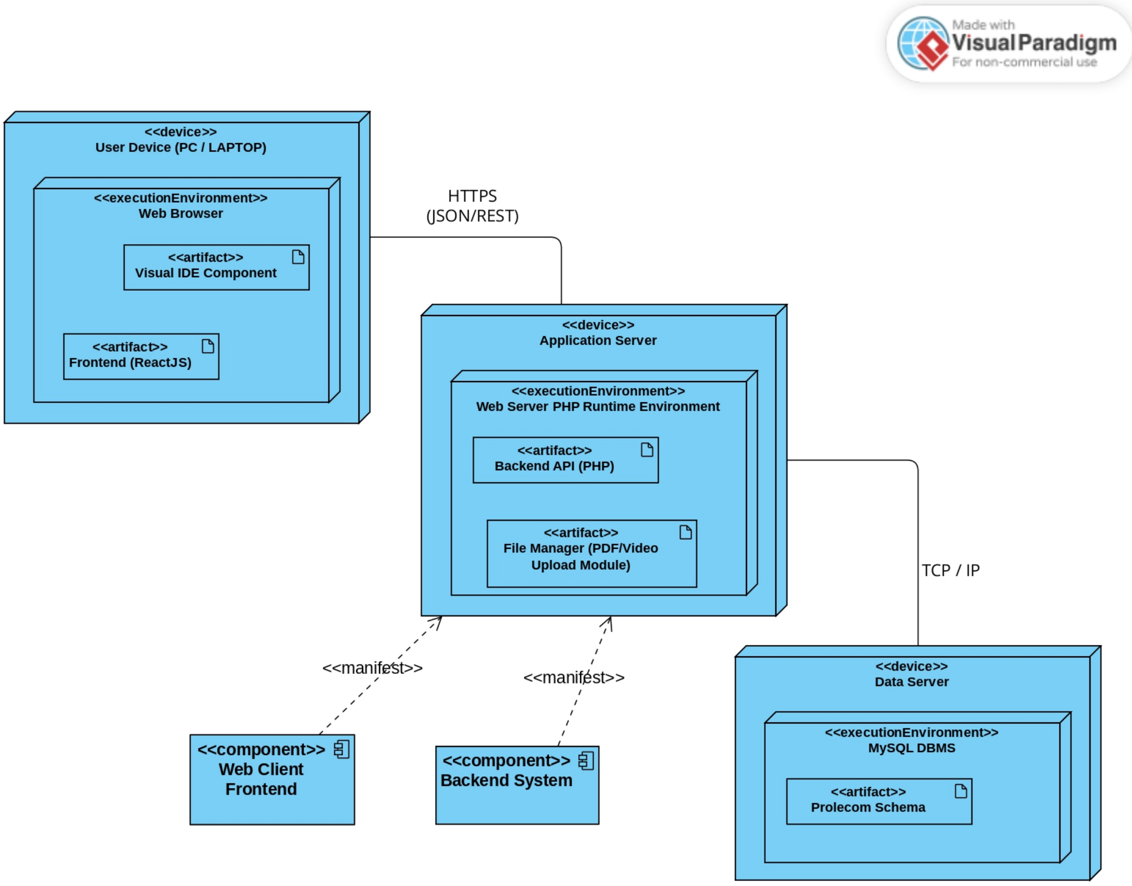

Figure 19: Web Module Deployment Diagram

Firma de Aprobación

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